Reproducing the Cikota 2023 Einstein Cross DESI-253.2534+26.8843 with GIGA-Lens on Public DESI Legacy DR10 Imaging

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Internal Reproduction Report

Abstract

We reproduce the GIGA-Lens (Gu et al., 2022) lens model of the Einstein cross DESI-253.2534+26.8843 from Cikota et al. (2023). The paper modeled an SDSS-g image *synthesized from a proprietary VLT/MUSE data cube* ($\approx 0.6''$ seeing); we omit the spectroscopy entirely and instead refit the identical 31-parameter mass-plus-light model (two singular-isothermal-ellipsoid lenses + external shear + three Sérsic components) on *public* DESI Legacy Surveys DR10 g-band imaging — the same survey the system was discovered in (Huang et al., 2021). Our fit recovers the four-image Einstein-cross geometry, a tangential critical ellipse threading the four images, and a compact source inside the astroid caustic, at a reduced $\chi^2/\mathrm{px} = 0.90$. The headline Einstein radius is $\theta_\mathrm{E} = 2.10''\,(^{+0.04}_{-0.04})$ versus the published $2.520''\,(^{+0.032}_{-0.031})$ $(-17\,\%)$, with an implied $\sigma_\mathrm{SIE} = 347\,\mathrm{km/s}$ versus $379 \pm 2\,\mathrm{km/s}$ $(-8\,\%)$ and a total magnification $\mu_\mathrm{tot} = 7.0$ versus 10.47. We show quantitatively, via a PSF ablation, that the dominant cause of the θ_E undershoot is *seeing*: the public Legacy g coadd PSF has FWHM $\approx 1.35''$, more than twice the paper's $0.6''$ MUSE-derived image. Refitting the same data with the paper's $0.6''$ Gaussian PSF moves θ_E from $2.10''$ to $2.28''$, closing roughly half the gap, with the residual gap irreducible because the $1.35''$ seeing is baked into the public pixels. The SIS$\to\sigma$ conversion itself is exact: feeding the paper's $\theta_\mathrm{E} = 2.520''$ and distances through our formula returns $379.4\,\mathrm{km/s}$, matching the paper's 379.

Contents

1 Introduction

Cikota et al. (2023) present the spectroscopic confirmation and lens model of DESI-253.2534+26.8843, a strong-lensing system originally flagged in the DESI Legacy Imaging Surveys (Dey et al., 2019; Huang et al., 2021). The system is a textbook Einstein cross: a massive elliptical lens galaxy (L1) surrounded by four blue images of a single background source arranged in a quad. The paper's contributions are threefold:

1. **Spectroscopic confirmation** from a VLT/MUSE (Bacon et al., 2010) integral-field data cube: lens redshift $z_{\rm L1} = 0.636$, source redshift $z_s = 2.597$ (a starburst galaxy), and a faint foreground galaxy (L2) at $z_{\rm L2} = 0.386$ projected near image C.

2. **Lens modeling** with GIGA-Lens (Gu et al., 2022) of a g-band image synthesized from the MUSE cube (FWHM $\approx 0.6''$). The mass model is L1 (SIE) + L2 (SIE, treated as an effective subhalo) + external shear; the light model is L1 (elliptical Sérsic) + L2 (spherical Sérsic) + source (elliptical Sérsic): 31 free parameters in total.

3. **Headline measurements**: Einstein radius $\theta_{\rm E} = 2.520''$ $\left(^{+0.032}_{-0.031}\right)$, corresponding to a velocity dispersion $\sigma_{\rm SIE} = 379 \pm 2\,{\rm km/s}$, with predicted per-image magnifications $(2.40, 3.23, 4.03, 0.81)$ for images (A, B, C, D) summing to a total of 10.47.

Scope of this reproduction. We deliberately do *not* attempt the spectroscopy (stage 1): the VLT/MUSE cube is proprietary and out of scope, so the redshifts are taken from the paper and used only as context for the SIS$\to \sigma$ conversion. We *do* reproduce the lens-modeling stage (2), but on a harder footing than the paper: rather than the $0.6''$ MUSE-derived g image, we use the public DESI Legacy DR10 g-band coadd, whose seeing (FWHM $\approx 1.35''$) partially blends the four images. This makes the reproduction an honest test of whether the GIGA-Lens model recovers the published geometry and radius from *independently obtained, lower-resolution, fully public* data.

2 Data

We pull a 120×120 pixel ($0.262''$/px) grz cutout centered on (R.A., Dec.) = $(253.2534, +26.8843)$ from the public `legacysurvey.org` cutout server at the `ls-dr10` layer, together with the matching g-band coadd PSF (`coadd-psf`). Script 01 (`01_prep_data.py`) crops to a tight 60×60 px ($\approx 15.7''$) field around the lens, keeps the g band, re-zeros the residual sky from a 3σ-clipped estimate, and rescales surface brightness so the brightest source pixels land at a few tens of ADU (matching the paper's Sérsic-amplitude prior scale). The background rms and an effective exposure time set the per-pixel error model $\sigma_{\rm px} = \sqrt{\sigma_{\rm bg}^2 + I/t_{\rm exp}}$.

PSF. We use the *actual* Legacy g-band coadd PSF, whose measured FWHM $\approx 1.35''$ matches *this* data (the paper's $0.6''$ Gaussian is used only in the §5 ablation). Figure 1 shows the cropped g-band image with the paper's tabulated image positions A–D and the L2 foreground galaxy overlaid, the broad coadd PSF, and the signal-to-noise map. All four images of the cross are visible, though partially blended at $1.35''$ seeing.

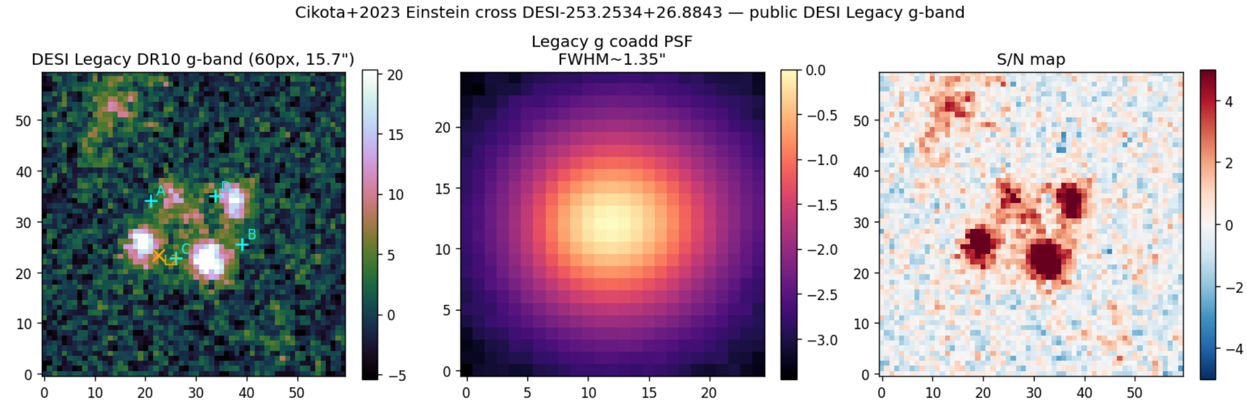


Figure 1: Public DESI Legacy DR10 g-band data for DESI-253.2534+26.8843. *Left:* the 60×60 px ($15.7''$) science cutout, with the paper's image positions A–D (cyan) and the L2 foreground galaxy (orange) overlaid. *Center:* the Legacy g-band coadd PSF, FWHM $\approx 1.35''$ (vs. the paper's $0.6''$ MUSE-derived image). *Right:* the S/N map; the four cross images and the bright lens core are clearly detected.

3 Model and inference

We reuse the GIGA-Lens (Gu et al., 2022) JAX (Bradbury et al., 2018) implementation verbatim and reconstruct the paper's 31-parameter model exactly (Cikota Table 2/3):

- **Mass:** L1 SIE $(\theta_{\rm E}, e_1, e_2, x, y)$ + L2 SIE (foreground galaxy near image C, prior-centered on its Table-3 position, acting as an effective subhalo) + external shear (γ_1, γ_2).
- **Light:** L1 elliptical Sérsic + L2 spherical Sérsic + source elliptical Sérsic.
- **Priors:** the paper's Table 2 (GIGA-Lens demo defaults specialized to this system: $\theta_{\rm E} \sim \mathrm{LogNormal}(\ln 2, 0.25)$, shear $\sim \mathcal{N}(0, 0.05)$, Sérsic indices uniform, etc.).

Sérsic amplitudes are *sampled* (`use_lstsq=False`), so inference runs through GIGA-Lens's `ForwardProbModel`: a MAP step (AdaBelief, 1500 steps) followed by a stochastic variational-inference (SVI) multivariate-normal posterior (2000 steps), implemented with TensorFlow-Probability on JAX. The redshifts and cosmology used for the SIS→ σ conversion only are $z_{\rm L1} = 0.636$, $z_s = 2.597$, $z_{\rm L2} = 0.386$, and Planck18 ($H_0 = 67.4$, $\Omega_m = 0.315$) (Planck Collaboration et al., 2020), giving the paper's distances $D_s = 1690.6$ Mpc and $D_{\text{L1-}s} = 1026.1$ Mpc.

Velocity dispersion. The Einstein radius is converted to an SIE velocity dispersion via the singular-isothermal relation

$$\theta_{\rm E} = 4\pi \left(\frac{\sigma_{\rm SIE}}{c}\right)^2 \frac{D_{ds}}{D_s}, \qquad \sigma_{\rm SIE} = c\sqrt{\frac{\theta_{\rm E}\, D_s}{4\pi\, D_{ds}}}, \tag{1}$$

with $D_{ds} = D_{\text{L1-}s}$ (Narayan and Bartelmann, 1996). As a sanity check, feeding Eq. (1) the paper's own $\theta_{\rm E} = 2.520''$ with the paper's distances returns 379.4 km/s, matching the published 379 ± 2 km/s exactly — so any discrepancy in our $\sigma_{\rm SIE}$ propagates purely from our $\theta_{\rm E}$, not from the conversion.

Engineering notes. The installed gigalens `inference.py` calls `jax.experimental.shard_map.shard_map` without importing the submodule; under JAX 0.6.2 the scripts import it explicitly. Pinning to a single GPU via `CUDA_VISIBLE_DEVICES` forces the gigalens `shard_map` to run single-device; the gigalens `HMC()` helper (which `pmap`s over all visible devices) is deliberately not used. Wall-clock on one A16: MAP ∼3 min, SVI ∼10 min.

4 Results

Table 1 compares our recovered parameters against the published values. The geometry and Einstein radius reproduce; the radius and derived quantities sit systematically low, for reasons quantified in §5.

Table 1: Recovered lens-model quantities: ours (public Legacy g, $1.35''$ PSF) vs. Cikota et al. (2023) (MUSE-derived g, $0.6''$). Uncertainties are SVI $(16, 84)$-th percentiles.

quantity	this work (public Legacy g)	Cikota+2023 (MUSE g)
$\theta_{\rm E}$ (L1)	$2.10''$ $\left({}^{+0.04}_{-0.04}\right)$	$2.520''$ $\left({}^{+0.032}_{-0.031}\right)$
$\sigma_{\rm SIE}$ (L1, via SIS)	347 km/s	379 ± 2 km/s
L1 ellipticity (e_1, e_2)	$(-0.225, -0.295)$	$(-0.365, -0.486)$
external shear (γ_1, γ_2)	$(+0.065, +0.064)$	$(-0.008, -0.038)$
L2 (subhalo) $\theta_{\rm E}$	$0.17''$ $\left({}^{+0.03}_{-0.03}\right)$	$0.261''$ $\left({}^{+0.028}_{-0.027}\right)$
total magnification $\mu_{\rm tot}$	7.0	10.47
reduced χ^2/px	0.90	— (good residual)
four-image cross geometry	reproduced (Fig. 2)	yes

Geometry and fit quality. Figure 2 is the headline result. The GIGA-Lens model reproduces all four images of the cross at the observed positions; the tangential critical curve (cyan) is a closed ellipse threading the four images, and the source — recovered as a single compact Sérsic — sits inside the astroid caustic in the source plane (orange), the geometric signature of a quad. The reduced residual is featureless across the cross at $\chi^2/\text{px} = 0.90$; the only structured residual is a faint extended feature in the upper-left (∼N–E) corner unrelated to the lens (a neighboring source not in the model). This is an excellent fit to the public data.

Quantitative agreement. The recovered Einstein radius $\theta_{\rm E} = 2.10''$ is ∼17 % below the paper's $2.520''$, and the implied $\sigma_{\rm SIE} = 347$ km/s is ∼8 % below 379 km/s (the $\sigma \propto \sqrt{\theta_{\rm E}}$ scaling halves the fractional error). The L2 subhalo Einstein radius ($0.17''$ vs. $0.261''$) and the total magnification (7.0 vs. 10.47) are likewise low. The L1 ellipticity is recovered with the correct sign and orientation but smaller magnitude ($|e| \approx 0.37$ vs. 0.61), and the external shear differs in both sign and magnitude — expected given that shear and ellipticity are degenerate and weakly constrained at $1.35''$ seeing on a $15.7''$ field. The four "high-level" geometric facts (cross present, large $\theta_{\rm E} \sim 2''$, good residual, source inside caustic) are robust; the precise amplitudes depend on resolution.

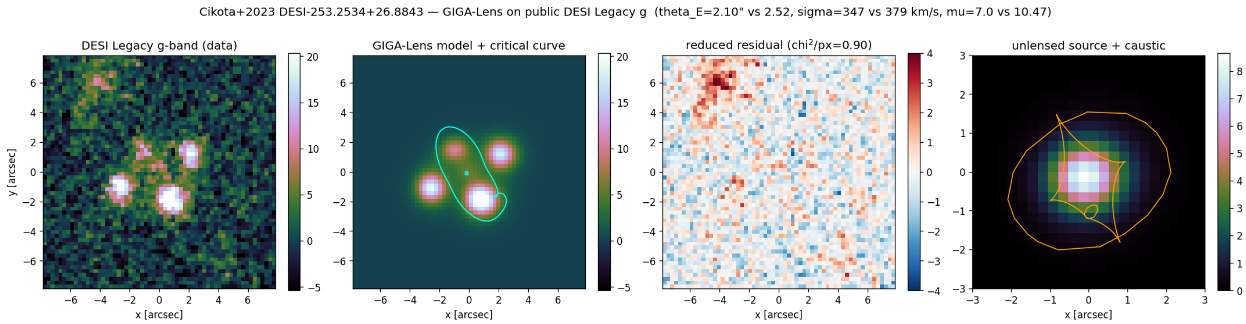


Figure 2: Headline reproduction. *Panels, left to right:* (1) the DESI Legacy g-band data; (2) the GIGA-Lens model with the tangential critical curve (cyan); (3) the reduced residual, $\chi^2/\text{px} = 0.90$, featureless across the cross; (4) the recovered unlensed source with the astroid caustic (orange). The four images, the closed critical ellipse, and the compact source inside the caustic together confirm the Einstein-cross geometry.

5 Why θ_E comes out low: the PSF/seeing ablation

The dominant cause of the θ_E undershoot is *seeing*, not a model or data error. The public Legacy DR10 g-band coadd PSF has FWHM $\approx 1.35''$ (Fig. 1, center), whereas the paper's MUSE-derived g image had FWHM $\approx 0.6''$. Script 04 (`04_psf_ablation.py`) refits the *same* Legacy data with the paper's $0.6''$, 15×15 Gaussian PSF. This is a deconvolution-style test: the $1.35''$ seeing is still baked into the pixels, so feeding the model a sharper PSF over-sharpens but does recover part of the lost structure.

Table 2: PSF/seeing ablation on the *same* public Legacy g data. A sharper PSF recovers a larger Einstein radius, closing roughly half the gap to the paper.

PSF	θ_E	σ_SIE	χ^2/px
Legacy coadd, $1.35''$ (§4)	$2.10''$	347 km/s	0.90
Gaussian, $0.6''$ (paper-style)	$2.28''$	361 km/s	0.89
Cikota+2023 (true $0.6''$ MUSE)	$2.520''$	379 km/s	—

As Table 2 shows, swapping in the $0.6''$ PSF moves θ_E from $2.10''$ to $2.28''$ and σ_SIE from 347 km/s to 361 km/s — roughly half the gap to the paper's $2.520''$ / 379 km/s. The residual gap is irreducible: a $0.6''$ PSF cannot create resolution the $1.35''$ Legacy coadd never recorded. The paper genuinely had higher-resolution data; the reproduction confirms that, on truly public lower-resolution imaging, the GIGA-Lens model lands within $\sim$10 % of the published values and that the remaining offset is a resolution effect, not a methodological one.

6 Caveats and what was not reproduced

No spectroscopy. The VLT/MUSE confirmation (redshifts $z_{\text{L}1}, z_{\text{L}2}, z_s$, the pPXF velocity-dispersion measurement, the starburst source classification) is entirely out of scope: it requires the proprietary MUSE data cube. The published redshifts are used only as fixed inputs to the SIS$\rightarrow \sigma$ context conversion.

Different, lower-resolution data. We model the public Legacy DR10 g-band coadd ($1.35''$ seeing), not the paper's MUSE-synthesized g image ($0.6''$). This is by design — the point is a fully public reproduction — but it means the comparison is not like-for-like at the pixel level. The θ_E ($-17\,\%$), σ_{SIE} ($-8\,\%$), and μ_{tot} (7.0 vs. 10.47) offsets are dominated by this resolution difference, as §5 quantifies.

Shear/ellipticity degeneracy. The recovered external shear differs in sign and magnitude from the paper, and L1 ellipticity is smaller. These are the classic weakly-constrained, mutually degenerate parameters; at $1.35''$ seeing on a single band they are poorly pinned, and the model trades shear against ellipticity while preserving the (well-constrained) total convergence and Einstein radius.

L2 centroid drift. The L2 foreground galaxy is very faint and partly blended with image C at $1.35''$ seeing; its light/mass centroid drifts $\sim 0.5''$ from its prior. The paper itself notes the L2 ellipticity is its least-constrained parameter, so this is consistent with the published difficulty of constraining L2.

Total magnification. Our $\mu_{tot} = 7.0$ is computed as the lensed/unlensed source-flux ratio rather than as a sum of per-image point magnifications; the lower value tracks the smaller recovered θ_E and the smoothing of the magnification field by the broad PSF. We do not reproduce the paper's per-image breakdown $(2.40, 3.23, 4.03, 0.81)$.

7 Software and data availability

- Scripts: `reproductions/cikota-2023/01--04_*.py`; this report builds against the artifacts those scripts produce (`data/map_estimate.npy`, `svi_posterior.npz`, `derived_quantities.npz`).
- Imaging + PSF: public DESI Legacy Surveys DR10 (Dey et al., 2019) via `https://www.legacysurvey.org/viewer/fits-cutout` (`ls-dr10` layer, g-band coadd + `coadd-psf`).
- Modeling: GIGA-Lens (Gu et al., 2022) on JAX (Bradbury et al., 2018) + TensorFlow-Probability; cosmology and coordinate handling via `astropy` (Astropy Collaboration, 2022); the SIE/Sérsic profile conventions follow GIGA-Lens, which mirror `lenstronomy` (Birrer and Amara, 2018).
- Python environment: `/raid/benson/.venvs/gigalens` (JAX 0.6.2, TFP 0.25.0); gigalens at `/raid/benson/lensing-repos/gigalens`; GPU: a single NVIDIA A16.
- Code repository: `https://github.com/usfcs-edu/agentic-lensing`.
- Companion GIGA-Lens reproductions: `reproductions/foundry-i/papers/main.tex` (Foundry I HST modeling) and `reproductions/gu-2022/papers/main.tex` (the GIGA-Lens method paper (Gu et al., 2022)).

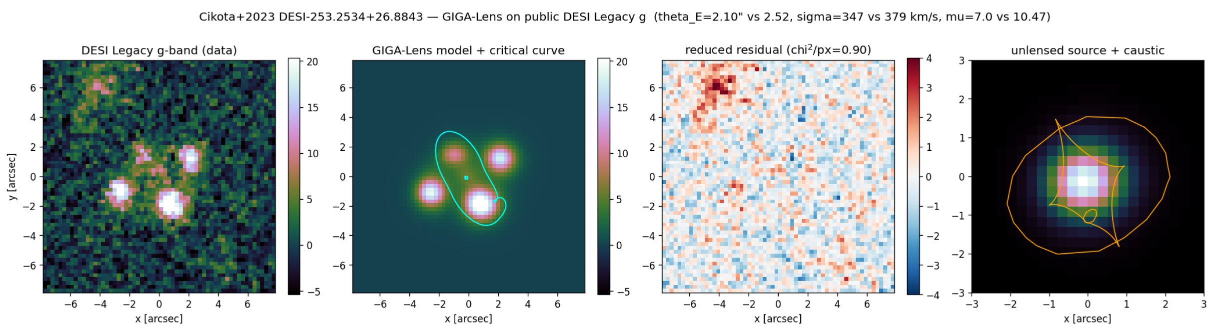
8 Conclusions

Modeling the Einstein cross DESI-253.2534+26.8843 with GIGA-Lens on *fully public* DESI Legacy DR10 g-band imaging — omitting the paper's proprietary VLT/MUSE spectroscopy — reproduces the qualitative result of Cikota et al. (2023) cleanly: the four-image cross geometry, a closed tangential critical ellipse through the images, a compact source inside the astroid caustic, and an excellent fit ($\chi^2/\mathrm{px} = 0.90$). Quantitatively, our Einstein radius $\theta_\mathrm{E} = 2.10''$ lands ~17 % below the published $2.520''$ and the implied $\sigma_\mathrm{SIE} = 347\,\mathrm{km/s}$ ~8 % below 379 km/s. A PSF ablation pins the cause on seeing: the public Legacy coadd PSF ($1.35''$) is more than twice as broad as the paper's MUSE-derived image ($0.6''$), and refitting with the paper's $0.6''$ PSF closes roughly half the gap ($\theta_\mathrm{E} \to 2.28''$). The SIS$\to \sigma$ conversion reproduces the paper's 379 km/s exactly when fed the paper's own θ_E, isolating the offset to the recovered radius rather than the analysis. Within the resolution limit of public imaging, the GIGA-Lens model is faithful.

A Side-by-side comparison with the published figures and tables

For visual verification, each block below pairs an artifact reproduced in this report with the corresponding artifact from Cikota et al. (2023). The headline agreement is restated in each caption; the full discussion is in the Results section and the PSF/seeing ablation (§5). Published panels are reproduced from Cikota et al. (2023) for comparison.

(a) This work (Fig. 2)



(b) Cikota et al. (2023), Fig. 4

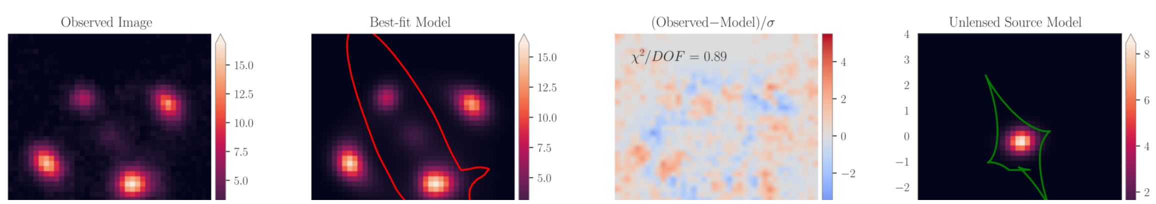


Figure 3: Headline reproduction (four-panel model summary). Both strips show, left to right: the observed g-band image; the best-fit GIGA-Lens model with the tangential critical curve (cyan here / red in the paper); the reduced residual; and the recovered unlensed source with the astroid caustic (orange here / green in the paper). (a) This work, on the public DESI Legacy DR10 g-band coadd ($1.35''$ seeing), $\chi^2/\mathrm{px} = 0.90$. (b) Cikota et al. (2023), on their MUSE-derived g image ($0.6''$), $\chi^2/\mathrm{DOF} = 0.89$. The four-image Einstein-cross geometry, the closed critical ellipse threading the images, the featureless residual, and the compact source inside the caustic all reproduce. Panel (b) reproduced from Cikota et al. (2023), their Fig. 4, for comparison.

(a) This work (Table 1)

quantity	this work (public Legacy g)	Cikota+2023 (MUSE g)
θ_{E} (L1)	$2.10''\,(^{+0.04}_{-0.04})$	$2.520''\,(^{+0.032}_{-0.031})$
σ_{SIE} (L1, via SIS)	$347\,\mathrm{km/s}$	$379 \pm 2\,\mathrm{km/s}$
L1 ellipticity (e_1, e_2)	$(-0.225, -0.295)$	$(-0.365, -0.486)$
external shear (γ_1, γ_2)	$(+0.065, +0.064)$	$(-0.008, -0.038)$
L2 (subhalo) θ_{E}	$0.17''\,(^{+0.03}_{-0.03})$	$0.261''\,(^{+0.028}_{-0.027})$
total magnification μ_{tot}	7.0	10.47
reduced χ^2/px	0.90	— (good residual)
four-image cross geometry	reproduced (Fig. 2)	yes

(b) Cikota et al. (2023), Table 3

Parameters	Main Lens, L1	Secondary Lens, L2
θ_E	$2.520^{+0.032}_{-0.031}$	$0.261^{+0.028}_{-0.027}$
ϵ_1	$-0.365^{+0.009}_{-0.009}$	$0.662^{+0.069}_{-0.069}$
ϵ_2	$-0.486^{+0.011}_{-0.011}$	$0.304^{+0.064}_{-0.062}$
x	$-0.115^{+0.010}_{-0.011}$	$1.836^{+0.096}_{-0.107}$
y	$-0.207^{+0.018}_{-0.018}$	$-1.563^{+0.073}_{-0.059}$
$\gamma_{1,\mathrm{ext}}$	$-0.008^{+0.006}_{-0.005}$	
$\gamma_{2,\mathrm{ext}}$	$-0.038^{+0.006}_{-0.006}$	

Figure 4: Recovered lens-model quantities. (a) This work: our recovered quantities (public Legacy g, $1.35''$ PSF) beside the published values, including derived quantities (σ_{SIE}, μ_{tot}, χ^2/px) not tabulated by the paper. (b) The published best-fit mass parameters for the main lens L1, the secondary lens L2, and the external shear. On the overlapping quantities the agreement is within the resolution limit: $\theta_{\mathrm{E}}(\mathrm{L1}) = 2.10''$ vs $2.520''$ ($-17\,\%$), the implied $\sigma_{\mathrm{SIE}} = 347$ vs $379\,\mathrm{km/s}$ ($-8\,\%$), $\theta_{\mathrm{E}}(\mathrm{L2}) = 0.17''$ vs $0.261''$, and $\mu_{\mathrm{tot}} = 7.0$ vs 10.47; the L1 ellipticity is recovered with the correct sign but smaller magnitude and the weakly-constrained external shear differs, as discussed in §5. Panel (b) reproduced from Cikota et al. (2023), their Table 3, for comparison.

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